

# NEWS RESEARCH ASSISTANT

Problem Definition · Data Processing · Evaluation Criteria

Capstone Project · Version 1.0 · 3 Members

## Part 1 — Problem Definition

### 1.1 Context & Motivation

Researchers, students, journalists, and knowledge workers regularly need to track fast-moving topics across multiple news sources. The current state-of-the-art involves two flawed extremes:

- Search engines return ranked links — the user must still open and read 10–15 articles to form a complete picture. This is slow, cognitively expensive, and loses provenance.
- General-purpose LLMs can synthesise text but hallucinate facts when used without grounding — especially for events after their training cutoff.

Neither approach alone is sufficient. The gap is a system that retrieves live, sourced articles and synthesises them into a grounded, cited answer automatically.

| Dimension       | Detail                                                                                                                                                 |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Domain          | News media consumption and information synthesis                                                                                                       |
| Affected Users  | Researchers, students, journalists, analysts, and knowledge workers                                                                                    |
| Core Pain Point | Manually reading 10–15 articles per query to synthesise a coherent answer is time-consuming and cognitively expensive                                  |
| Existing Gap    | Search engines return ranked links, not synthesised answers with provenance. LLMs hallucinate without grounding. Neither solution alone is sufficient. |
| Proposed Answer | An agentic RAG system that ingests live news, retrieves relevant chunks, and generates cited, multi-source summaries — automatically.                  |

### 1.2 Problem Statement

Given a natural-language research question, the system must:

1. Retrieve the most relevant recent news articles from a continuously updated corpus.

2. Synthesise a concise, accurate answer that is fully grounded in the retrieved articles.

3. Cite every factual claim with source name, date, and URL.

4. Refuse to answer — without hallucinating — when no relevant articles exist in the knowledge base.

### 1.3 Functional & Non-Functional Requirements

| ID   | Type           | Requirement                            | Acceptance Condition                                                            |
|------|----------------|----------------------------------------|---------------------------------------------------------------------------------|
| R-01 | Functional     | Ingest articles from RSS and NewsAPI   | Pipeline pulls and stores articles every 6–12 hrs without manual intervention   |
| R-02 | Functional     | Retrieve relevant chunks for any query | Cosine similarity search returns k=5–8 chunks with Recall@5 > 0.70              |
| R-03 | Functional     | Generate cited, multi-source answer    | Output includes inline [Source, Date] citations; faithfulness score > 4.0/5     |
| R-04 | Functional     | Handle thin retrieval gracefully       | Agent triggers broaden_search when < 3 chunks found; broaden trigger rate > 80% |
| R-05 | Functional     | Refuse unanswerable queries honestly   | Returns 'Not in knowledge base' without hallucinating; graceful refusal > 90%   |
| R-06 | Non-Functional | Recency of cited sources               | Average article age in cited answers < 3 days                                   |
| R-07 | Non-Functional | Source diversity per answer            | Minimum 2 unique news sources cited per answer                                  |
| R-08 | Non-Functional | Agent reasoning correctness            | Tool call accuracy > 85%; hallucination rate = 0%                               |
| R-09 | Non-Functional | System runs fully local                | ChromaDB + Streamlit deployable on a single developer machine, no cloud infra   |

### 1.4 Scope & Constraints

| Constraint     | Details                                                                                     |
|----------------|---------------------------------------------------------------------------------------------|
| Time           | 7 days, 3 team members — no fine-tuning, no auth systems, no cloud deployment required      |
| Data           | Public news only via free-tier NewsAPI and RSS feeds; no paywalled sources                  |
| Infrastructure | All vector storage and UI run locally (ChromaDB + Streamlit); no managed services           |
| LLM Access     | Claude claude-sonnet-4-20250514 or GPT-4o-mini via API — pay-per-use, no dedicated instance |
| Scope          | Technology news domain only (TechCrunch, Wired, The Verge) for initial demo                 |

## Part 2 — Data Processing

### 2.1 Ingestion & Processing Pipeline

Articles move through seven sequential stages from raw web source to queryable vector entries. The pipeline runs on a scheduled loop; all stages are idempotent.

| Stage | Name      | Tools                                      | Output / Notes                                                                        |
|-------|-----------|--------------------------------------------|---------------------------------------------------------------------------------------|
| 1     | Fetch     | feedparser, NewsAPI, newspaper3k           | Raw HTML articles pulled from RSS feeds + API; runs on APScheduler every 6–12 hrs     |
| 2     | Clean     | BeautifulSoup, newspaper3k                 | Strips HTML tags, ads, nav bars, footers, and boilerplate; extracts clean body text   |
| 3     | Dedupe    | MD5 hash on URL + title                    | Drops duplicate articles before embedding to avoid redundant vector entries           |
| 4     | Chunk     | LangChain RecursiveCharacterText Splitter  | 300–500 token segments, 50-token overlap; preserves sentence boundaries               |
| 5     | Embed     | text-embedding-3-small or all-MiniLM-L6-v2 | Each chunk converted to a dense vector; same model used for queries at retrieval time |
| 6     | Store     | ChromaDB (local)                           | Vector + metadata stored: article_id, title, source, date, url, chunk_index           |
| 7     | Freshness | APScheduler + metadata age check           | Articles older than 30 days soft-deleted; 'last 7 days' filter available via MCP      |

### 2.2 Vector Store Schema

Each chunk stored in ChromaDB carries the following metadata fields. Metadata is used by MCP filter tools and displayed in the citation output.

| Field       | Type          | Description                                                            |
|-------------|---------------|------------------------------------------------------------------------|
| article_id  | string (UUID) | Unique identifier for the article; used by fetch_full_article MCP tool |
| title       | string        | Article headline; shown in citations and UI source panel               |
| source      | string        | Publisher name (e.g. TechCrunch); used by filter_by_source tool        |
| date        | ISO 8601 date | Publication date; used by filter_by_date tool and recency eval metric  |
| url         | string (URL)  | Canonical article URL; included in final cited answer output           |
| chunk_index | integer       | Position of this chunk within the full article; aids context ordering  |
| embedding   | float[]       | Dense vector (1536-dim for text-embedding-3-small, 384-dim for MiniLM) |
| chunk_text  | string        | Raw text of this segment; included in the LLM context prompt           |

## 2.3 Chunking & Embedding Design Decisions

| Decision                                      | Rationale                                                                                                                                  |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 300–500 token chunk size                      | Large enough to contain a complete thought or claim; small enough to stay within LLM context limits and maintain retrieval precision       |
| 50-token overlap                              | Prevents important content at chunk boundaries from being missed during retrieval; overlap is modest to avoid excessive vector duplication |
| RecursiveCharacterSplitter                    | Respects paragraph and sentence boundaries before falling back to character splits, preserving semantic coherence within chunks            |
| Same embedding model at ingest and query time | Vector space alignment is critical — a mismatch between embedding models at write and read time degrades similarity scores                 |
| Top-k deduplication                           | Keep only the highest-scoring chunk per source article in the final context to maximise source diversity in answers                        |

## 2.4 Data Quality Controls

### Deduplication

Articles are fingerprinted by MD5 hash of URL + title before embedding. Duplicates (reposts, syndicated content) are dropped at Stage 3 to prevent inflated retrieval scores.

### Freshness management

Articles older than 30 days are soft-deleted from ChromaDB. The filter\_by\_date MCP tool lets the agent apply a tighter window (e.g. last 3 days) at query time when the user asks for 'latest' or 'today'.

### Cleaning fidelity

BeautifulSoup and newspaper3k remove structural noise. After cleaning, chunks below 50 tokens (typically navigation fragments) are discarded before embedding to avoid low-quality vector entries.

### Embedding model consistency

The same embedding model must be used at both ingestion and query time. Switching models requires a full re-embed of the ChromaDB collection — this is logged as a migration task.

# Part 3 — Evaluation Criteria

## 3.1 Evaluation Philosophy

The eval harness tests four independent failure modes, each with its own layer. A system can score well on retrieval but poorly on answer quality — layered evals expose this. All metrics are automated except citation accuracy, which requires a 20% manual spot-check.

- Layer 1 — Retrieval: did the right articles surface?
- Layer 2 — Answer Quality: is the generated answer faithful and relevant? (LLM-as-judge)
- Layer 3 — Source Quality: are citations accurate, diverse, and recent?
- Layer 4 — Agent Behaviour: did the agent reason and act correctly?

## 3.2 Full Evaluation Matrix

| Layer | Metric            | Category  | Target     | Definition & Measurement                                                                                           |
|-------|-------------------|-----------|------------|--------------------------------------------------------------------------------------------------------------------|
| L1    | Recall@k          | Retrieval | Pass >0.70 | Were all relevant articles present in the top-k results? Manually label relevant IDs per query and check coverage. |
| L1    | Precision@k       | Retrieval | Pass >0.60 | What fraction of returned chunks were actually relevant? Count relevant / k, average across test set.              |
| L1    | MRR               | Retrieval | Pass >0.65 | Mean Reciprocal Rank — was the best article ranked highest? Rewards high-rank relevant results.                    |
| L1    | NDCG@k            | Retrieval | Pass >0.65 | Normalised discounted cumulative gain — position-weighted relevance score.                                         |
| L2    | Faithfulness      | Answer    | Pass >4.0  | Every claim in the answer is directly supported by a retrieved source article (LLM-as-judge, 1–5 scale).           |
| L2    | Relevance         | Answer    | Improve    | Answer directly addresses the user's actual question (LLM-as-judge, 1–5 scale). Current baseline: 3.7.             |
| L2    | Completeness      | Answer    | Pass >3.5  | All key points from retrieved chunks are reflected in the answer (LLM-as-judge).                                   |
| L2    | Conciseness       | Answer    | Pass >3.5  | Answer is appropriately brief without losing key information (LLM-as-judge).                                       |
| L3    | Source Diversity  | Citations | Pass ≥2    | Number of unique news sources cited per answer. Prevents single-source echo-chamber summaries.                     |
| L3    | Recency Score     | Citations | Pass <3d   | Average age (in days) of cited articles. Ensures answers reflect current events.                                   |
| L3    | Citation Accuracy | Citations | Pass >90%  | Does the cited source actually contain the claimed fact? Manual spot-check across 20% of answers.                  |
| L3    | Coverage Gap Rate | Citations | Pass >85%  | Correct 'Not in knowledge base'                                                                                    |

| Layer | Metric             | Category | Target    | Definition & Measurement                                                                                    |
|-------|--------------------|----------|-----------|-------------------------------------------------------------------------------------------------------------|
|       |                    |          |           | responses for genuinely unanswerable queries; no false citations.                                           |
| L4    | Tool Call Accuracy | Agent    | Pass >85% | Did the agent call the right tools (and in the right order) for each query type?                            |
| L4    | Avg Tool Calls     | Agent    | Pass 2–4  | Average number of tool calls per successful answer. Too few suggests skipped steps; too many wastes tokens. |
| L4    | Broaden Trigger    | Agent    | Pass >80% | How often broaden_search was correctly triggered when initial retrieval returned fewer than 3 chunks.       |
| L4    | Hallucination Rate | Agent    | Pass 0%   | Answers generated without first calling search_news. Must be exactly 0% — hard quality gate.                |
| L4    | Graceful Refusal   | Agent    | Pass >90% | Correct 'Not in knowledge base' responses for out-of-scope queries; no hallucinated citations.              |

### 3.3 LLM-as-Judge Prompt (Layer 2)

Answer quality metrics (faithfulness, relevance, completeness, conciseness) are evaluated by calling the LLM with the following structured prompt. Scores are collected on a 1–5 scale and averaged across the 30-query test set.

```
Score this answer on FAITHFULNESS (1-5):
  5 = Every claim directly supported by a source article.
  3 = Most claims supported; 1-2 unsupported additions.
  1 = Answer largely fabricated or contradicts sources.

Question: {query}
Answer:  {answer}
Articles: {chunks}

Return JSON: { "score": <1-5>, "reason": "<string>" }
```

### 3.4 Test Dataset Design

A 30-query evaluation dataset is manually curated across five query categories to ensure broad coverage:

- Standard factual queries — 10 queries where relevant articles exist and retrieval should succeed straightforwardly.
- Recency-sensitive queries — 6 queries containing 'latest', 'today', or 'this week'; tests filter\_by\_date triggering.
- Thin-retrieval queries — 6 queries on niche sub-topics likely to return < 3 initial chunks; tests broaden\_search.
- Multi-hop queries — 4 queries where the first answer naturally leads to a follow-up retrieval step.
- Out-of-scope queries — 4 queries on topics entirely absent from the knowledge base; tests graceful refusal.

### 3.5 Baseline Scorecard

The following scorecard reflects performance on the 30-query test set after the initial system integration. Metrics marked IMPROVE are prioritised for Day 6 remediation.

| Metric             | Score | Target | Status  | Interpretation                         |
|--------------------|-------|--------|---------|----------------------------------------|
| Recall@5           | 0.76  | > 0.70 | PASS    | Above threshold — retrieval is sound   |
| Precision@5        | 0.63  | > 0.60 | PASS    | 63% of returned chunks were relevant   |
| Faithfulness (avg) | 4.2/5 | > 4.0  | PASS    | Answers stay grounded in sources       |
| Relevance (avg)    | 3.7/5 | > 4.0  | IMPROVE | Below target — prompt tuning needed    |
| Source Diversity   | 2.4   | >= 2   | PASS    | Average 2.4 unique sources per answer  |
| Recency Score      | 1.9 d | < 3 d  | PASS    | Articles cited avg 1.9 days old        |
| Tool Call Accuracy | 88%   | > 85%  | PASS    | Right tools called 88% of the time     |
| Hallucination Rate | 0%    | 0%     | PASS    | Hard gate — zero violations            |
| Graceful Refusal   | 93%   | > 90%  | PASS    | Out-of-scope queries handled correctly |

### 3.6 Remediation Plan

#### Relevance score (currently 3.7 / 5 — below 4.0 target)

1. Review the 6 queries where relevance scored < 3. Identify whether the issue is retrieval (wrong chunks returned) or generation (answer drifts from the question).
2. If retrieval-side: tune the embedding model or increase k and apply rerank\_chunks.
3. If generation-side: refine the generate\_summary system prompt to explicitly instruct the model to stay focused on the question.
4. Re-run the 30-query eval after each change. Target improvement to > 4.0 by end of Day 6.